Decompression DOE in Plastic Injection Molding

Technical Manual for Cold Runner and Hot Runner Production Systems

Prepared as a shop-floor and engineering reference for process engineers, technicians, tooling engineers, quality engineers, and maintenance personnel.

Scope: reciprocating-screw thermoplastic injection molding, including cold runner molds, hot runner molds, thermal gates, valve gates, nozzle shutoffs, and residence/degradation interactions.

Important truth-in-use note: this document is technically sourced and structured for professional review, but it has not been independently peer-reviewed by named outside experts. A formal peer-review checklist is included so a plant can validate it against its machines, materials, molds, customer requirements, and safety procedures before release as controlled work instruction.

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Executive Summary

A decompression DOE is a controlled experiment that varies screw pullback amount, timing, and speed to find the minimum effective decompression window that relieves residual melt pressure without introducing new instability. In most plants the setting is called decompression, suck-back, pull-back, or screw retraction after recovery. Some controls also allow decompression before recovery, before injection, or both.

The central tradeoff is simple: too little decompression leaves pressure in the barrel/nozzle/runner system and produces drool, gate stringing, cold slugs, and inconsistent startup; too much decompression can draw air into the melt stream, destabilize the check ring, reduce shot-to-shot repeatability, cause splay or blisters, and create false evidence of degradation.

Cold runner and hot runner molds must not be treated as equivalent. In a cold runner, the sprue and runner freeze and leave with the shot, so decompression troubleshooting usually centers on the machine nozzle, sprue, runner, cold slug, gate vestige, and first-shot fill. In a hot runner, the manifold and drops remain hot and pressurized between cycles, so decompression interacts with manifold balance, hot-tip thermal stability, valve-gate timing, material residence, stagnant pockets, color change, and degradation risk.

The repeated wording in the original requirement around “Residence/degradation study” is important but not identical to a Decompression DOE. A residence/degradation study intentionally measures material damage as a function of time, temperature, moisture, shear, barrel size, and runner/hot-runner hold-up volume. A Decompression DOE may reveal symptoms that look like degradation, but it does not by itself quantify molecular-weight loss. The correct plant approach is to separate the two studies and then map the interaction between them.

Critical distinction	Decompression DOE	Residence/degradation study	Why the distinction matters
Primary question	How much, when, and how fast should the screw retract to relieve pressure?	How much material damage occurs from thermal/shear history and dwell time?	Solving drool with excessive pullback can create air/splay and mislead the degradation investigation.
Main factors	Post- and pre-decompression amount, decompression speed, timing, nozzle/hot runner condition.	Barrel capacity, shot size, cycle time, melt temperature, screw rpm, back pressure, hot-runner volume, moisture.	Hot runners add hold-up volume that cold runners do not.
Main responses	Drool, stringing, cushion, part weight, first-shot fill, splay, gate quality, pressure curve.	MFR/MVR shift, color shift, black specks, odor, mechanical performance, molecular weight proxy tests.	A stable decompression window should be locked before judging residence effects.
Best output	Lowest effective setting with a validated process window.	Maximum safe dwell/temperature/shear envelope and purge/startup rules.	Both outputs become standard process controls.

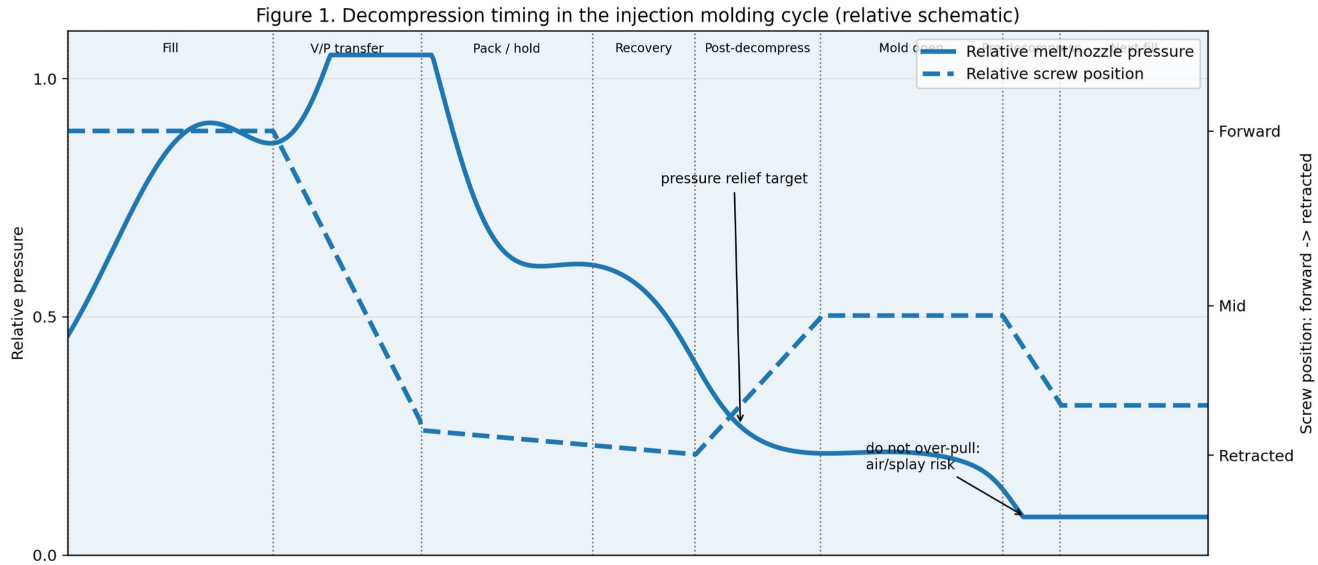

Figure 1. Relative cycle timing showing where decompression can occur. Actual controls and phase names vary by machine manufacturer.

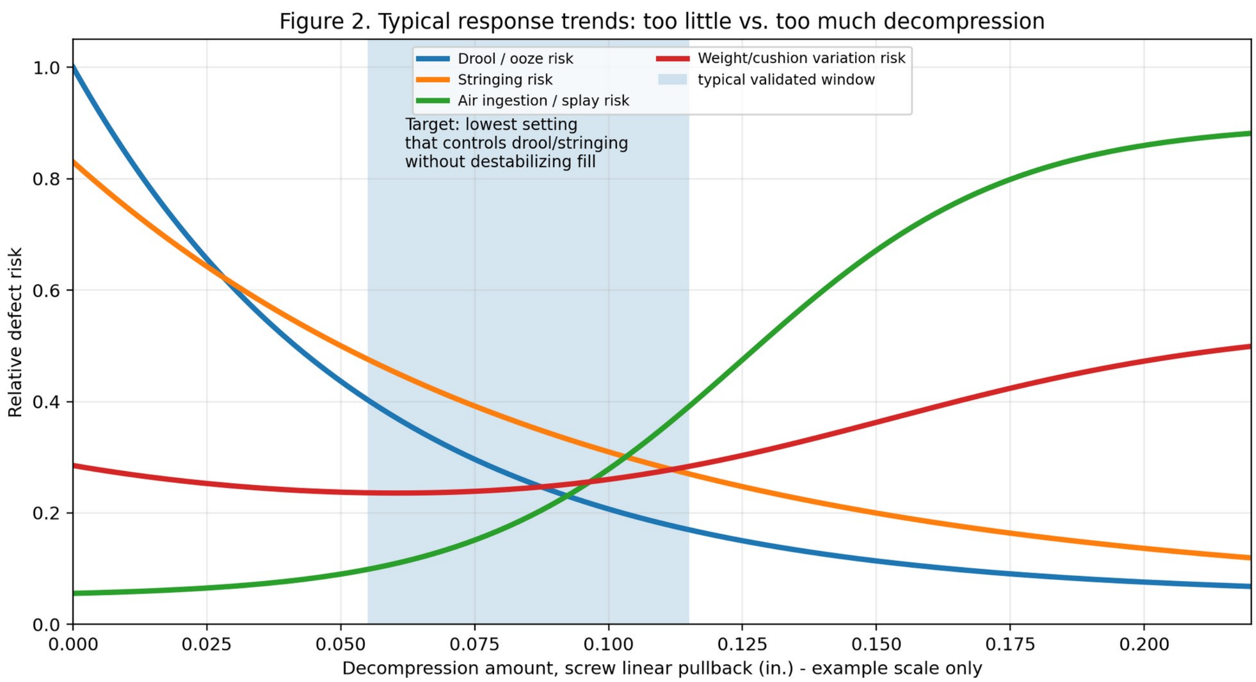

Figure 2. Typical response trends. The optimum is not maximum decompression; it is the lowest setting that controls pressure-driven defects without causing air, splay, short shots, or instability.

1. Definition and Overview

1.1 Precise technical definition

A Decompression DOE is a structured design of experiments used to determine the effective screw retraction strategy required to reduce residual melt pressure at the front of the screw, machine nozzle, sprue, runner, hot-tip, or hot-runner feed system after plasticizing or before the next injection event. The independent variables are decompression amount, decompression speed, decompression timing, and sometimes decompression mode. The dependent variables are pressure-relief symptoms and process-stability responses, including drool length, stringing, part weight, cushion repeatability, first-shot fullness, recovery time, splay, blisters, black specks, odor, color shift, gate quality, and pressure-curve repeatability.

In mechanical terms, decompression is screw movement backward without rotation. Moldex3D describes suck-back as the screw moving backward after reaching metering position, creating additional space, moving melt at the nozzle tip backward, dropping pressure, and increasing melt specific volume [4]. Resin and processing guides commonly caution that only small or minimum decompression should be used when needed because excessive pullback can draw air into the nozzle and create splay, blisters, or other defects [2][3].

1.2 What a successful outcome looks like

- No measurable nozzle drool at the production mold-open interval and carriage/nozzle state.
- No sprue, gate, or hot-tip stringing at normal mold-open speed and robot/sprue-picker timing.
- No first-shot shortness after normal cycling, brief interruption, or validated startup sequence.
- Part weight is stable across the DOE replicate set and across cavities. Use part-weight range, standard deviation, and Cpk/Ppk where the customer requires statistical release.
- Cushion at transfer/end-of-hold remains repeatable. An optimum that causes cushion drift is not an optimum.
- No new air hooks, splay, blisters, voids, black specks, smoke, odor, or color shift.
- Hot runner molds maintain cavity balance, gate vestige control, and hot-tip/manifold thermal stability.
- The setting repeats after material lot change, color change, normal purge, mold stop, and restart within the defined process window.

1.3 Observable failure outcomes

Condition	Too little decompression	Too much decompression	Common false conclusion
Machine nozzle / cold sprue	Drool, long strings, cold slug entering next shot, buildup on sprue bushing.	Air drawn through nozzle, startup hesitation, short first shot, splay near gate.	Assuming all splay is moisture instead of checking decompression amount and nozzle seat.
Cold runner gate	Sprue/gate stringing, heavy vestige, delayed break-off, visible tail.	Gate freeze instability, incomplete first fill, bubble/silver streak close to gate.	Blaming gate design before verifying the nozzle and decompression window.
Hot runner thermal gate	Gate drool, smears, hot-tip leakage, stringing.	Tip freeze-off, air draw, cavity imbalance, short cavities, black specks from oxygen/thermal cycling.	Assuming a manifold temperature issue when decompression created a pressure/air problem.
Valve-gated hot runner	Leakage past pin or pressure marks when valve closes under excess residual pressure.	Loss of pressure behind closed system, inconsistent first fill, delayed pressure recovery.	Assuming valve timing is wrong without checking pressure relief and gate seal.
Shot consistency	Higher pressure at next injection, inconsistent transfer, screw bounce/pressure decay.	Low or unstable cushion, weight spread, pressure curve drift.	Calling the check ring bad without separating NRV leakage from decompression effect.

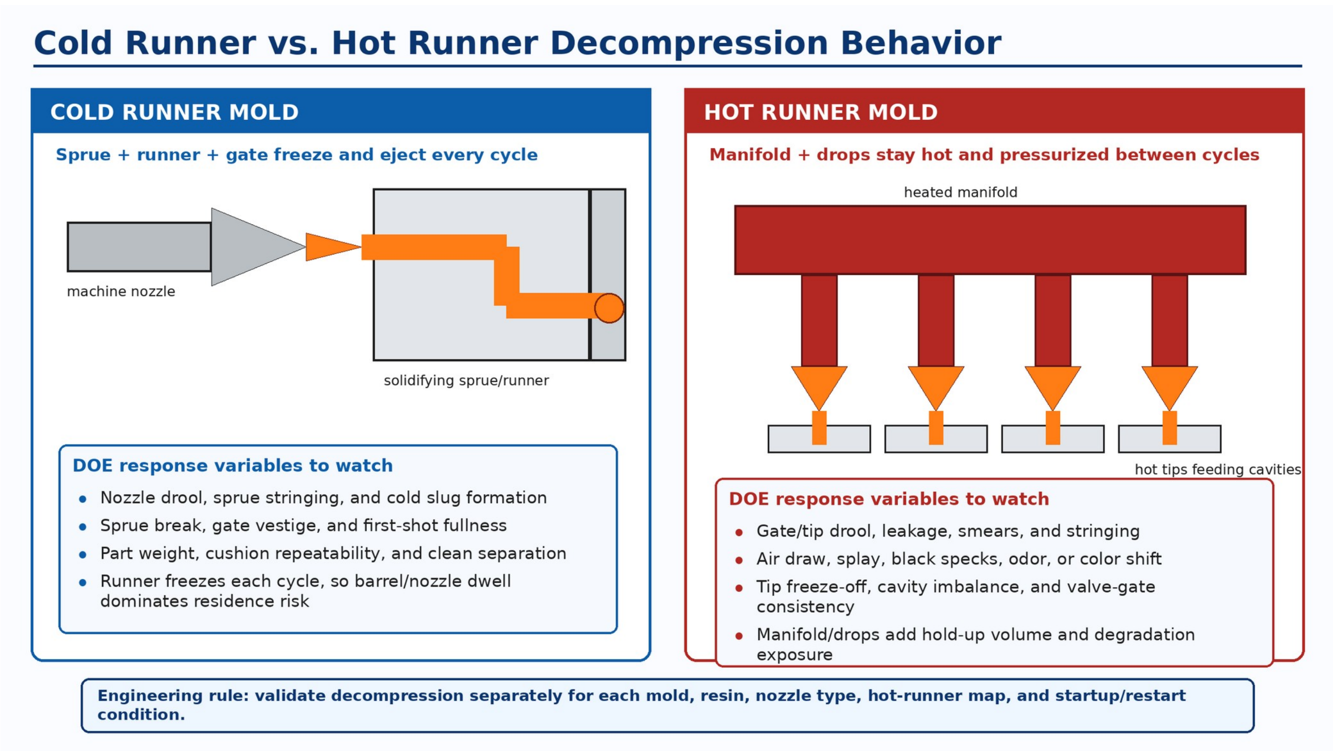

Figure 3. Cold runner and hot runner systems respond differently because the cold runner freezes each shot while the hot runner remains molten and thermally active.

2. Why a Decompression DOE is Done: Root Causes and Failure Modes

A decompression DOE is justified when production symptoms indicate residual melt pressure, nozzle/hot-tip leakage, pressure decay instability, or interaction between pressure relief and material health. It is not a generic first response for every cosmetic defect. It should be run after baseline checks confirm the mold, machine, material, and auxiliary equipment are capable of stable operation.

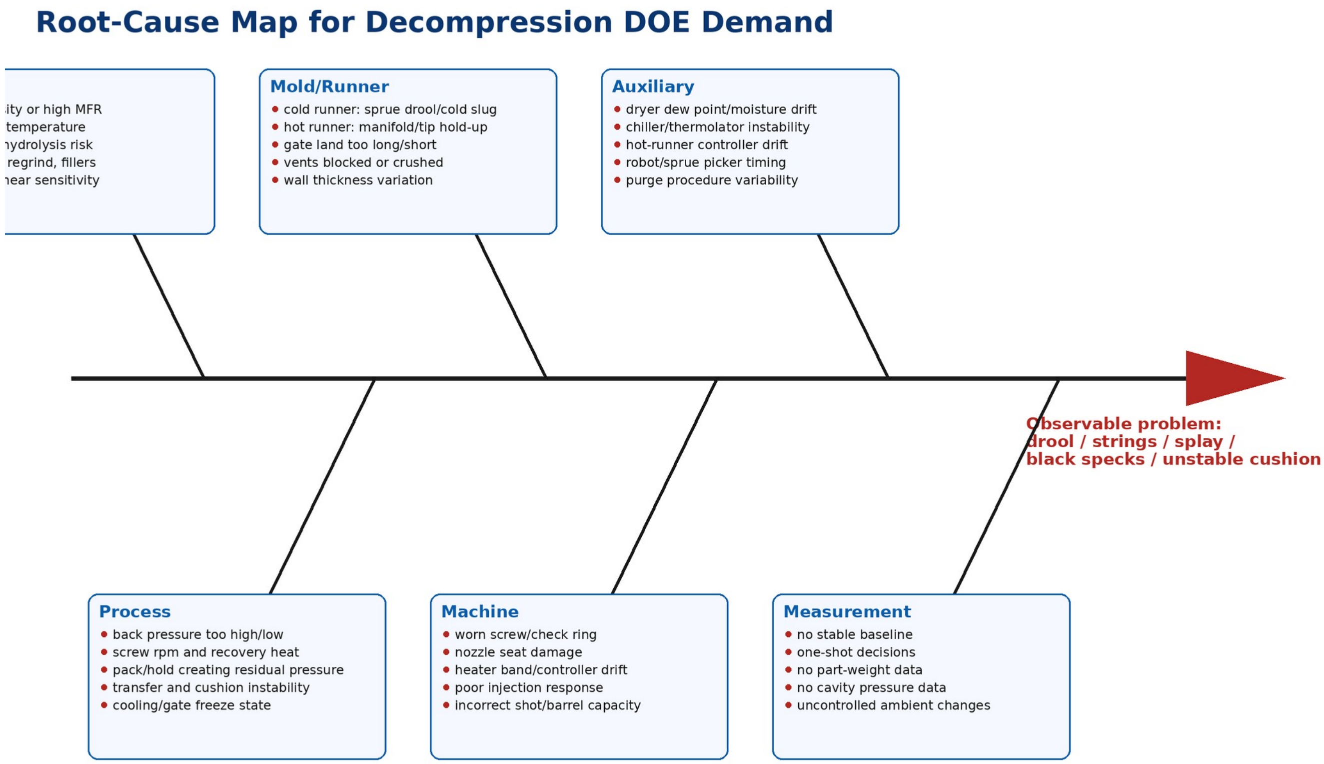

Figure 4. Fishbone-style root-cause map. Decompression is one lever, not the whole root-cause universe.

2.1 Material factors

Material behavior changes the required decompression window because melt viscosity, compressibility, thermal conductivity, crystallization behavior, moisture sensitivity, shear sensitivity, filler loading, and additives determine whether pressure can relax naturally or remains trapped long enough to cause drool and stringing. The same screw pullback that is harmless in a forgiving polyolefin may create splay, hydrolysis, or brittle performance in a moisture-sensitive engineering resin.

Material / family	Expected decompression behavior	Residence/degradation sensitivity	Practical interpretation
PP / PE polyolefins	Often relatively forgiving; low-viscosity grades and high MFR grades can drool/string more readily. Fast injection is commonly used when cosmetics allow.	Generally not hydrolysis-driven, but excessive melt temperature, long residence time, regrind, contamination, high screw speed, and high back pressure can cause color change, black specks, or property loss.	Do not over-pull to solve stringing. First verify nozzle temperature, back pressure, gate freeze, hot-tip temperature, and material MFR.
ABS	Moderate viscosity; moisture can create surface defects; high shear and overheating can discolor or degrade but behavior is grade-dependent.	Moisture often causes cosmetic splay rather than backbone hydrolysis, but heat/shear/residence can still damage material.	If splay appears as decompression increases, separate air ingestion from dryer performance before changing drying specs.
PC	Higher melt temperature and high viscosity; sensitive to moisture-induced hydrolysis and thermal exposure.	Under-dried PC can degrade severely, particularly at elevated melt temperature and long residence; degradation may show MFR shift, brittleness, color shift, and silver streaks.	Keep decompression conservative. Validate dryness and melt temperature before concluding that decompression caused all visual issues.
PC/ABS	Often processed for cosmetics; stringing is common in public practitioner reports.	Blend can show color shift, streaking, moisture-related defects, and degradation if overheated or held too long.	Small changes in nozzle temperature, decompression, back pressure, and gate condition can matter.
POM, PA, PBT, PET, TPU	Behavior depends strongly on grade and moisture control.	Several are hydrolysis-sensitive or moisture-sensitive; drying and residence control are high priority.	Use resin supplier drying, temperature, and purge recommendations as hard constraints.

PVC / CPVC / heat-sensitive compounds	Usually avoid high shear and long dwell; drool solutions must not increase degradation risk.	Very heat/shear-sensitive; degraded rigid PVC can yellow and generate HCl odor according to processing guides [16].	A decompression DOE must be subordinate to strict thermal/shear limits and ventilation/EHS procedures.
Filled / glass / mineral grades	Higher viscosity and abrasive flow; gates and hot tips may act differently.	Fiber residence can affect length distribution; fillers can increase shear heating and wear.	Watch screw/check-ring wear. Cavity balance may change as decompression alters pressure recovery.

Material composition and additives matter. Lubricants, mold release, slip agents, color concentrates, flame retardants, glass fibers, minerals, impact modifiers, recycled content, and carrier resin compatibility can all change apparent viscosity and degradation symptoms. A decompression DOE must record complete material traceability: resin grade, lot, colorant lot, regrind percentage, dryer settings, measured moisture where required, purge history, and time at temperature.

2.2 How different polymers change residence/degradation study results

- PP and PE: degradation is usually driven by oxidation, high melt temperature, shear, residence, contamination, and regrind history rather than hydrolysis. Manufacturer guides for PE/PP list black specks and brittleness among symptoms linked to excessive melt temperature/residence in barrel or runner systems, excessive back pressure, excessive screw speed, and material hanging up in screw, check ring, or hot runner [15].
- ABS: moisture can create splay; high melt temperature, long residence, and shear history can produce discoloration and property changes. ABS grades vary widely, so supplier temperature and drying limits override generic ranges.
- PC: moisture, melt temperature, and residence time are critical. Covestro explains that moisture can cause surface streaks/bubbles and, in hydrolysis-sensitive plastics, molecular-chain degradation that may not be visible externally but can cause premature brittle failure [8]. AIM Institute reports that increasing melt temperature, residence time, and moisture increases degradation risk and includes PC/PET study data showing large MFR shifts in wet/high-temperature/long-residence conditions [13].
- Semi-crystalline vs amorphous behavior: semi-crystalline polymers may set up faster at the mold wall and can be sensitive to cooling and crystallization; amorphous materials often show clearer stress, splay, and optical defects. Mold temperature affects frozen-layer growth and therefore pressure loss, shear heating, and perceived flow stability [9].
- Hot runner effects: hot runners add continuous molten hold-up. A resin that survives barrel residence in a cold runner tool may degrade in a hot runner if manifold volume, tip dead spots, color-change dead legs, or zone drift extend effective residence time.

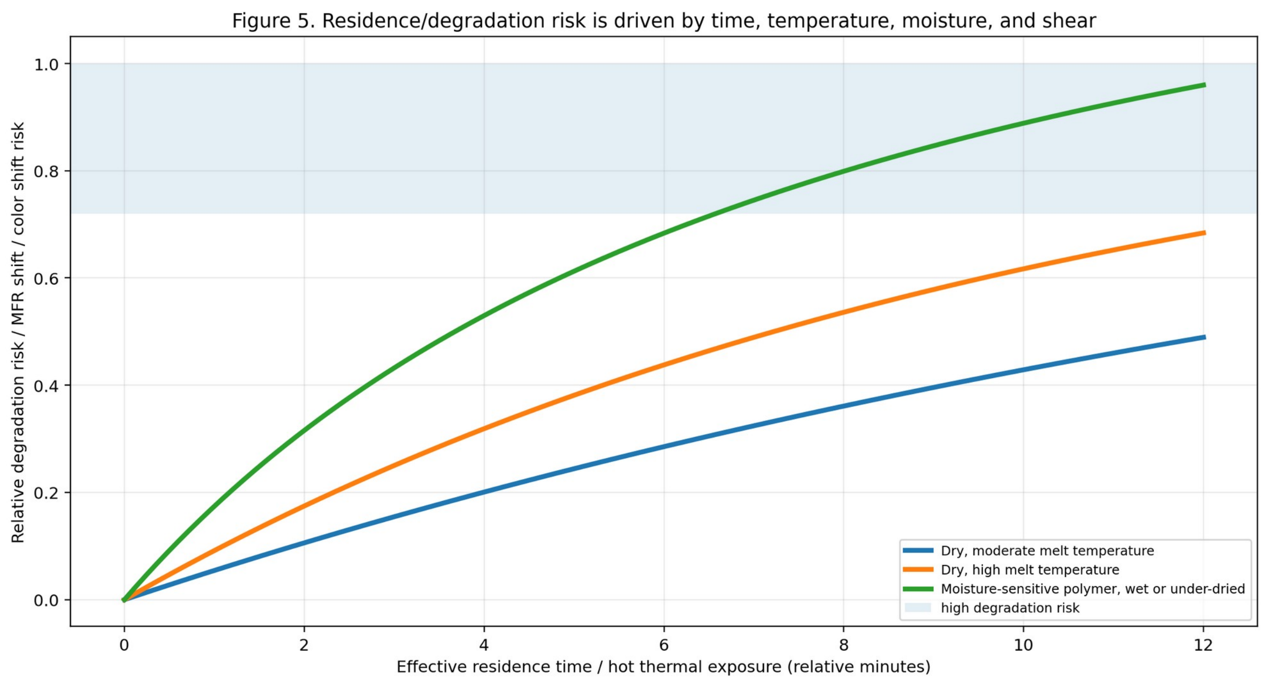

Figure 5. Residence/degradation study behavior is not the same as decompression response, but the studies interact strongly.

2.3 Process parameters

Parameter	Effect on decompression DOE	Cold runner emphasis	Hot runner emphasis
Injection speed	Changes pressure demand, shear heating, fill balance, and whether strings are pulled before gate freeze. High speed can reduce viscosity through shear but can also create shear defects.	Verify short-shot fill pattern and gate freeze before altering decompression. Cold slug wells and sprue behavior matter.	High speed may amplify hot-tip shear, gate blush, and cavity imbalance. Monitor pressure and gate quality.

Injection pressure / limit	A low pressure limit can masquerade as decompression-induced shortness. A high limit can hide high-viscosity or plugged-flow problems.	Hold injection pressure limit constant during DOE unless safety requires otherwise.	Do not use decompression to compensate for an imbalanced manifold or undersized tips.
Transfer position / V-P switchover	Directly affects cushion and pressure at end of fill. If transfer drifts, decompression conclusions are invalid.	Confirm 95-99% first-stage fill target where applicable before DOE.	Cavity pressure transfer can reveal actual melt behavior and compensate for nonreturn-valve effects [18].
Hold pressure/time	Creates residual pressure until gate freeze. Excess hold before gate freeze can increase drool after gate/sprue separation.	Run gate-freeze study first if uncertain.	Valve-gate timing and manifold pressure decay must be evaluated together.
Back pressure	Raises melt density, mixing, shear heating, and pressure in front of screw. Too high can aggravate drooling; too low can reduce melt homogeneity.	Hold constant during DOE. Reduce only if validated as root cause.	Back pressure plus manifold compression can make hot runner pressure relief nonlinear.
Screw rpm / recovery	Higher rpm increases shear heat and can reduce recovery time but may increase degradation risk and pressure variability.	Record recovery time and torque/load trend.	Hot runner systems can mask barrel instability until gate balance drifts.
Melt temperature profile	Lower viscosity raises drool tendency but improves flow. High temperature can increase degradation risk.	Nozzle temperature is high-leverage for sprue strings and cold slugs.	Hot-tip and manifold zones are high-leverage; one drifting zone can look like decompression failure.
Mold temperature / cooling	Changes frozen layer, gate freeze, shrinkage, pressure loss, and string break behavior.	Cold runner freezes each shot; temperature controller stability affects sprue/gate break.	Hot runner is less affected in manifold but gate area and cavity cooling still control gate freeze and vestige.

Covestro process-variable guidance emphasizes that during filling, frozen layers form near the cold mold wall and their thickness depends on injection time, melt temperature, and mold wall temperature. Shear stress and internal friction are greatest between the frozen layer and center of the flow channel, which explains why injection speed, mold temperature, gate geometry, and wall thickness can change both pressure response and thermal stress [9].

2.4 Mold design: cold runner vs. hot runner

Design feature	Cold runner impact	Hot runner impact	DOE control / observation
Sprue and runner	Solidify every cycle; excessive residual pressure causes sprue drool, strings, and cold slugs. Runners add shot size but not continuous molten residence after ejection.	Not applicable except hot sprue/hybrid systems.	Record sprue break, cold slug well performance, runner balance, and sprue-puller timing.
Manifold and drops	Not present.	Remain molten; add hold-up volume, temperature-control risk, material residence, and dead-spot degradation risk.	Document every zone setpoint/actual, startup soak time, color history, and leak/gate balance.
Thermal gate / hot tip	Cold gate freezes; gate land and mold temperature determine string break.	Tip heat and gate land determine drool, freeze-off, splay, and vestige.	Observe gate with the part still in mold when safe. Compare cavities.
Valve gate	Rare in cold runners but possible in hybrid systems.	Pin timing and seal condition can mechanically stop flow, but pressure behind the gate still matters.	Lock valve sequence while DOE runs unless valve timing is itself a studied factor.
Gate size and land	Small gate increases shear and gate freeze; large gate can stay open longer and string.	Tip/gate size affects pressure balance and residence in drops.	Avoid using decompression to hide bad gate design beyond validated limits.
Wall thickness variation	Thick sections stay molten longer and influence sink/pack; thin sections freeze and raise pressure.	Same part behavior, plus hot-runner gate area may overheat or underheat.	Short-shot and pressure studies identify hesitation and flow imbalance before DOE.
Venting and air traps	Blocked vents produce burns, short shots, and false decompression conclusions. Excessive clamp can close vents.	Same, plus hot-runner imbalance can change air-trap location by cavity.	Clean vents and verify vent depth/land condition before finalizing setting.

2.5 Machine factors

Machine factor	How it can distort DOE	Required check before DOE
Screw design	Compression ratio, L/D, flight depth, mixing section, and check-ring design affect melt density, shear heating, and pressure recovery. A screw that is wrong for the resin can make decompression appear more important than it is.	Confirm screw type and condition match resin family, filler, shot size, and required recovery.
Screw/check-ring wear	A worn nonreturn valve leaks during injection and creates cushion/weight drift that can be	Run an NRV/check-ring repeatability check. Compare cushion, transfer pressure, and weight

	mistaken for decompression error.	across at least 10 shots.
Barrel/nozzle heater control	Temperature drift changes viscosity, drool, and cold slug behavior. A bad nozzle heater can cause both drool and freeze-off cycling.	Verify actual temperature with safe external checks and controller history; inspect bands and thermocouple seating.
Back pressure and hydraulic/electric response	Actual back pressure may oscillate; electric and hydraulic response differ. Slow decompression response can leave pressure at the wrong time.	Trend recovery time, screw position, decompression end, and any machine alarms.
Shot size / barrel capacity	A very small shot in a large barrel increases residence time and thermal exposure; a large shot can reduce plasticizing margin.	Target an appropriate shot-size-to-barrel-capacity window per material supplier guidance. Lubrizol, for example, recommends considering runner/sprue and notes 60-75% barrel capacity as desirable for some TPU processing to minimize residence while preserving flow [14].
Clamp force	Clamp force does not directly decompress melt, but it changes vent function, flash risk, mold deflection, parting-line leakage, and cavity pressure. Too much clamp can close vents; too little can flash.	Set clamp using a validated method and hold it constant during DOE unless clamp is a separate study factor.
Nozzle seat and shutoff nozzle	Damaged seats, contamination, or shutoff mechanisms can create stringing/drool or trap material. Public practitioner discussions warn that material trapped between check valve and shutoff can degrade, and that melt decompression can be preferred for drool control in some cases [21].	Inspect nozzle radius/orifice compatibility, nozzle tip condition, shutoff actuation, and leakage path.

2.6 Auxiliary equipment

Auxiliary equipment	Role in decompression DOE	Failure mode that can cause false conclusion
Dryer / material handling	Controls moisture and feed consistency. Moisture-sensitive materials cannot be evaluated accurately if dryer dew point, temperature, or residence is unstable.	Moisture splay blamed on decompression; degraded PC/PBT/PET/TPU blamed on hot runner.
Chiller	Stabilizes mold cooling and sometimes feed throat/oil systems. Cooling drift changes gate freeze and part dimensions.	Stringing or first-shot changes blamed on decompression when gate freeze changed.
Thermolator / mold temperature controller	Controls mold wall temperature, frozen-layer growth, gate freeze, and dimensional stability.	Burns/short shots/flow lines blamed on decompression when mold temp is drifting.
Hot-runner temperature controller	Controls manifold and tip viscosity, pressure retention, gate freeze, and material residence. This is critical in every hot runner decompression DOE.	One hot or cold zone creates cavity imbalance, drool, black specks, or shortness wrongly attributed to decompression.
Robot / sprue picker	Changes timing of mold open, part removal, sprue break, and string observation.	String length appears to improve/worsen because takeout timing changed.
Grinder / regrind feed	Changes bulk density, fines content, moisture, and thermal history.	Shot-size variation or degradation blamed on decompression.
Purge equipment / purging compound	Determines how well degraded material is removed from screw, check ring, nozzle, hot tips, and manifold.	Black specks persist after DOE because degraded hang-up was never removed.

3. DOE Design Methodology and Data Collection

A decompression DOE must be small enough to run on the shop floor but disciplined enough to avoid false conclusions. The most common failure is changing decompression while temperature, moisture, screw recovery, hot-runner zones, robot timing, and material feed are still drifting. Another common failure is judging only the drool response and ignoring first-shot fullness, weight stability, splay, and degradation symptoms.

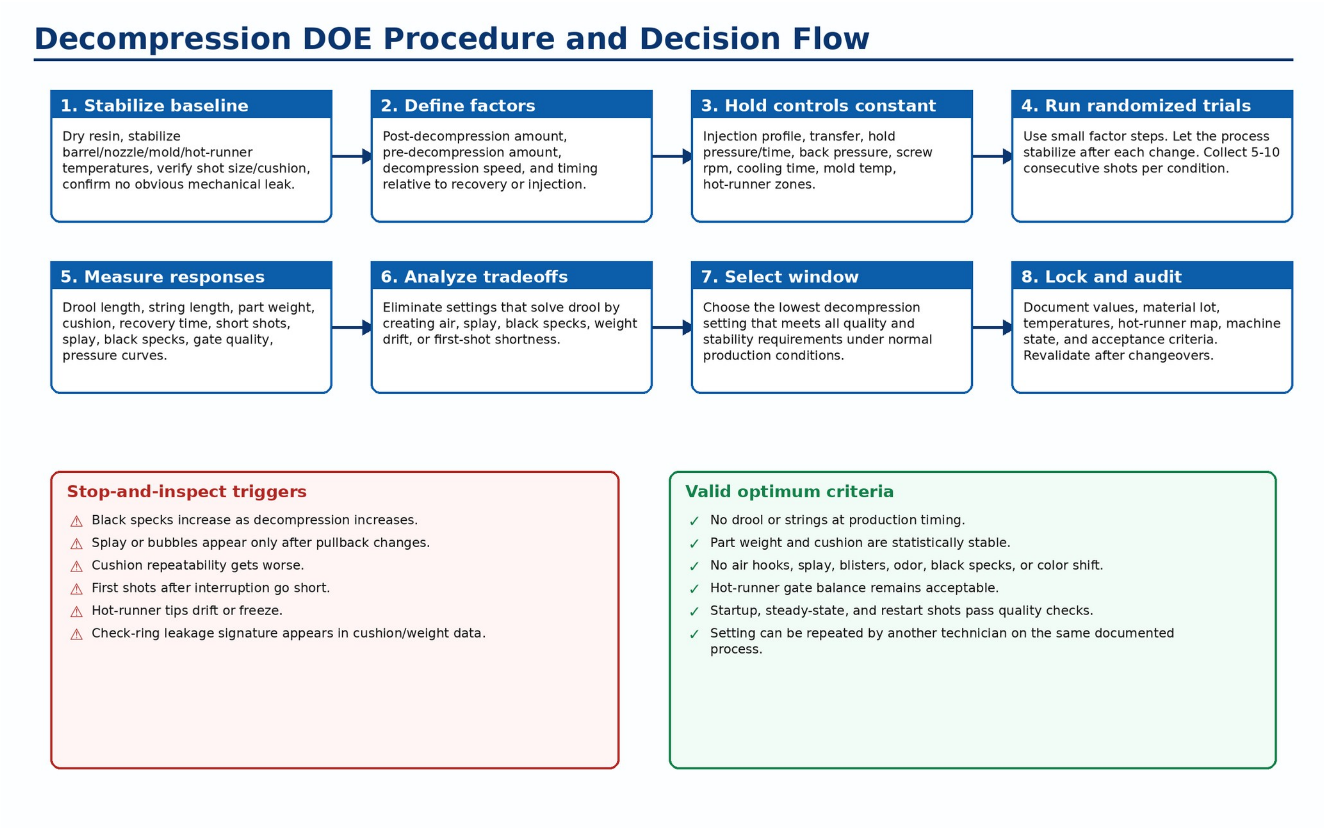

Figure 7. DOE procedure and decision flow for safe decompression optimization.

3.1 Preconditions before testing

- The mold is clean, vents are open, water circuits are connected correctly, and no obvious gate damage, hot-tip damage, or sprue-bushing damage is present.
- Material is correct, dry where required, within approved regrind percentage, and traceable by lot.
- Barrel, nozzle, mold, and hot-runner temperatures are stable at production setpoints. Hot runners require actual-zone verification, not just setpoint review.
- Shot size, cushion, transfer, hold pressure/time, cooling time, back pressure, screw rpm, and injection speed profile are locked unless deliberately studied.
- No machine alarms are active, including decompression-end, charge-end, heater, motor load, mold protection, or hot-runner controller alarms.
- The nonreturn valve/check ring is repeatable enough for a DOE. If weight and cushion are already drifting at baseline, fix the injection unit first.
- Safe observation method is defined. Never place hands between platens or near purge/nozzle discharge. Follow lockout/tagout and plant safety rules for inspection.

3.2 Factor selection

Factor	Definition	Typical starting strategy	Notes
Post-decompression amount	Screw retraction after screw recovery/plasticizing.	Start at current approved setting or a small value; increase in small increments.	Most common factor. Values are machine-, screw-, resin-, and process-specific.
Pre-decompression amount	Screw retraction before next injection, or before/after recovery depending on control terminology.	Use only when process need is clear; many tools do not require it.	Can affect first-shot fill and pressure recovery more aggressively than post-decompression.
Decompression speed	Rate of screw retraction during pullback.	Prefer controlled/slow-to-moderate pullback if air ingestion is suspected.	Very fast pullback can pull air and destabilize melt near nozzle/tip.
Timing / sequence	When decompression occurs relative to recovery, mold open, valve-gate closure, carriage retract, or injection start.	Document exactly as machine control names it.	Do not compare presses unless the control sequence is equivalent.
Mode / pressure-based control	Some systems allow decompression to position, volume, pressure, or machine-interface setting.	Use stable repeatable mode first.	Advanced pressure-based approaches must be validated for shot repeatability.

3.3 Response variables to measure

Response	How to measure	Unit / scale	Acceptance logic
Drool length	Measure melt tail from nozzle/sprue/hot-tip at a defined time after mold open or carriage movement. Use photos with scale.	mm or in	Minimize, but not at expense of splay/weight/cushion.
String length	Measure longest string between gate/sprue/nozzle and part/runner at peak occurrence.	mm or in	Minimize; record cavity and gate location.
Part weight	Weigh 5-10 consecutive shots; multi-cavity tools require per-cavity or family weight strategy.	g	Range and standard deviation must remain within quality plan.
Cushion	Read at transfer, end of hold, and end of recovery where controller supports it.	mm or in	Must be repeatable; drifting cushion invalidates DOE.
First-shot fullness	Run after standardized interruption/startup.	% full or pass/fail	No short first shot after normal production interruption.
Splay / blister / air hooks	Visual inspection under controlled lighting; severity 0-5.	rating	Any new air defect as decompression increases is a hard warning.
Black specks / color shift / odor	Visual, colorimeter, particle count, odor note, purge sample, or lab analysis.	rating / delta E / count	In hot runners, this may signal material residence, oxygen, or dead spots.
Pressure curve	Machine injection pressure or cavity pressure where available.	bar/psi/MPa vs time	Look for repeatability and pressure recovery; not just peak pressure.
Gate quality	Gate vestige height, smear, string, freeze-off, blush.	mm/rating	Hot runners require per-cavity and per-drop review.

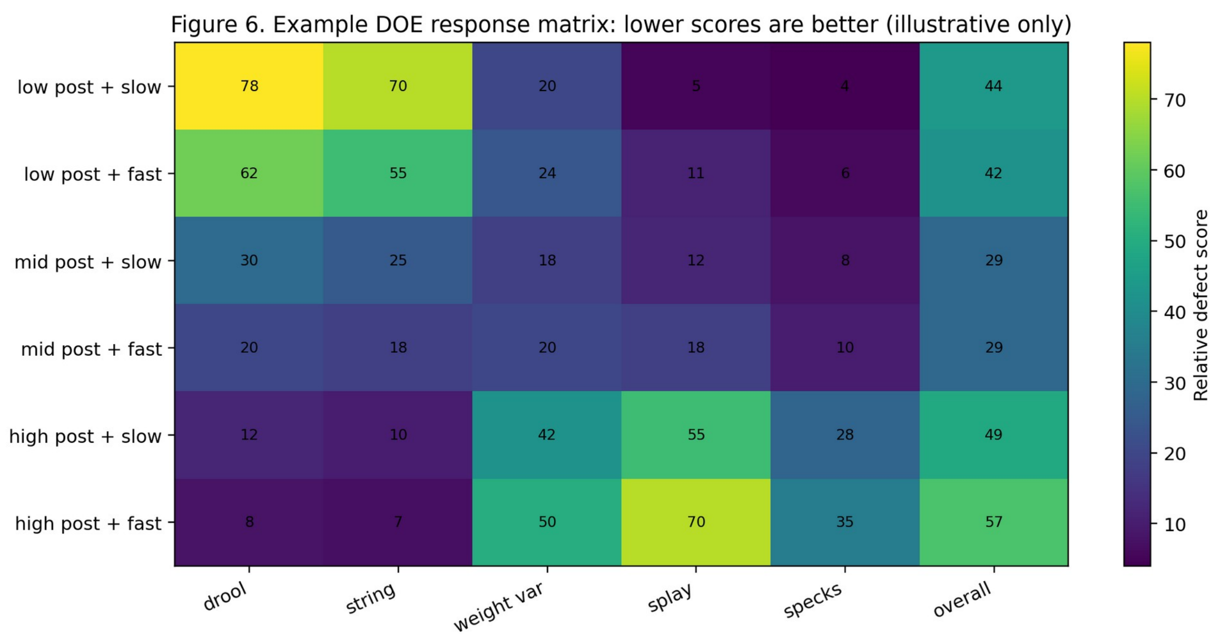

Figure 8. Example result matrix. Numerical values are illustrative; use plant-specific acceptance thresholds.

3.4 Recommended experimental designs

Design	Use case	Pros	Limits
One-factor-at-a-time screening	Troubleshooting a production issue with limited downtime.	Simple; technician-friendly; fast.	Can miss interactions between speed, amount, hot-runner zones, and pre/post timing.
2-level factorial screening	New tool launch or chronic defect. Factors: post amount, speed, pre amount on/off, timing.	Finds main effects and interactions with fewer trials than full random trial-and-error.	Requires discipline and randomization; still may need confirmation runs.
Response surface / center points	High-value tools, medical/automotive validations, hot-runner systems with tight windows.	Maps optimum window and curvature; better for robustness.	More trials and analysis; requires stable baseline.
Sequential DOE	Start broad, then narrow around optimum.	Best practical approach for most plants.	Must document decision rules before moving to next round.
Split cold/hot validation	Molds that can run cold-runner family, hot sprue, hot runner, or stack variants.	Prevents transferring cold-runner settings into hot-runner process.	More setup work, but avoids high-risk false assumptions.

3.5 Example screening matrix

Run	Post-decompression amount	Decompression speed	Pre-decompression	Order note	Required observations
1	Low	Slow	Off	Randomized	Drool, string, weight, cushion, visual, pressure.
2	High	Slow	Off	Randomized	Same. Stabilize several cycles before recording.
3	Low	Fast	Off	Randomized	Same; watch for air/splay.
4	High	Fast	Off	Randomized	Same; high risk of air if pullback excessive.
5	Low	Slow	On, small	Randomized	Same; compare first-shot fill.
6	High	Slow	On, small	Randomized	Same; hot runner: monitor gate balance.
7	Low	Fast	On, small	Randomized	Same; reject if splay or shortness appears.
8	High	Fast	On, small	Randomized	Same; likely upper-risk boundary.
9	Center point	Moderate	Off or small	Repeat 2-3 times	Estimate process noise and repeatability.

4. Real-World Examples and Case Studies

The following cases are written for training use. Where they are based on public practitioner reports, they are identified as such. Where they are composite examples, they are labeled as field-style scenarios rather than claimed as proprietary interviews. The purpose is to teach diagnostic logic, not to imply that one universal adjustment solves every plant problem.

Case	System	Observed symptom	Study performed	Resolution logic
Public practitioner report: PC/ABS strings	Machine nozzle / gate interface; ABS, PC, PC/ABS context.	A molder reported hardened strings hanging from the nozzle when the gate pulled away; attempted nozzle temperature, back pressure, screw rpm, tip type, and mold temperature changes [20].	A decompression DOE would isolate decompression amount/speed/timing while holding temperature and back pressure constant, then check part weight and splay.	Stringing may improve with decompression, but the final window must prove no short shots, splay, or degradation symptoms.
Public forum discussion: shutoff nozzle pressure loss	Machine with shutoff nozzle.	Second shot had apparent pressure/fill difficulty; practitioner warned that trapped hot material between check valve and shutoff may degrade and noted melt decompression as a drool-control option [21].	Separate shutoff function check from decompression DOE; test nozzle heating, mechanical actuation, and trapped-material dwell.	A shutoff nozzle is not automatically superior to decompression; it can add dead volume and degradation risk.
Composite medical closure case	32-cavity PP child-resistant closure, hot runner valve gate.	Intermittent gate strings and black specks after weekend startups.	Residence/degradation study plus decompression DOE. Factors: hot-runner soak time, purge volume, post-decompression amount, valve-gate close timing.	Root cause was not only decompression. Hot tips were overheated during idle; final controls included lower idle temperature, purge/startup sequence, smaller pullback, and cavity-specific gate audit.
Composite automotive bracket case	Glass-filled PA hot runner.	Field failures; internal voids suspected.	Part weight, cavity pressure, CT scan, and residence review. FORCE Technology documents a similar value proposition for CT: full 3D data and identifiable voids in injection molded brackets [17].	Use CT/NDT and mechanical testing for critical structural parts; decompression DOE alone cannot verify structural integrity.
Composite consumer-goods cold runner case	ABS cold runner with tunnel gates.	Sprue strings and cold slugs entering the next shot.	Post-decompression amount and speed DOE after nozzle heater verification and gate-freeze study.	A small increase in post-decompression plus nozzle-temperature correction eliminated strings. Excess pullback was rejected because it produced silver streaks

				near the gate.

4.1 Technician and engineer experience patterns

- Experienced technicians often first look for timing: whether the string forms at mold open, sprue break, carriage retract, robot takeout, or purge. The timing tells whether the symptom is nozzle pressure, gate freeze, hot-tip leakage, or mechanical takeout.
- Engineers tend to over-focus on the numeric setting and under-focus on the sequence. Two machines can both show “0.080 in decompression” while executing different timing and speed profiles.
- Hot-runner technicians routinely treat zone stability as a first-class variable. A decompression DOE is weak if it ignores actual manifold and tip zone behavior.
- Quality engineers should insist that the accepted setting be proven with part weight, visual criteria, and restart shots. A setting that only looks clean for five steady-state cycles is not robust.
- Maintenance should be pulled in early when cushion repeatability is poor. Worn check rings, nozzle seats, damaged tips, leaking shutoffs, or heater-band drift can invalidate every DOE run.

5. Diagnostic Techniques and Verification Methods

5.1 Can you identify after production is running that a DOE was actually done?

No visual inspection method can prove that a decompression DOE or residence/degradation study was performed. Parts can show evidence of a stable or unstable process, but the existence of a DOE is proven only by controlled records: setup sheets, machine data logs, trial matrix, raw measurements, photos, material lot records, hot-runner actuals, and documented decision rules. A clean-running process with no drool/stringing does not prove a DOE; it only proves the current condition is acceptable at the time observed.

- Evidence that a DOE likely occurred: locked decompression values in the setup sheet, response table with measured drool/string/weight/cushion, dated approval by process/quality/tooling, baseline and confirmation runs, and notes on cold vs hot runner differences.
- Evidence that no credible DOE occurred: only a final setting with no measurements, no material/hot-runner records, no replicate shots, no cold/hot distinction, no restart validation, or uncontrolled changes during trial.
- Part evidence of poor or missing optimization: drool residue, recurring gate strings, random splay after pullback changes, unexplained short first shots, black specks after idle, large weight spread, or cavity-to-cavity hot-runner drift.

Inspection and Diagnostic Methods for Decompression-Related Defects

Method	Best used for	Limitations / controls
Visual inspection	Drool, strings, cold slugs, gate vestige, splay, specks, color shift	Fast; operator-friendly; subjective without lighting standard
Part weight + cushion run chart	Shot-to-shot stability and first-shot response	Low cost; requires disciplined sampling
Cavity pressure / temp sensors	Fill balance, transfer repeatability, short-shot prediction	Best production feedback; requires mold instrumentation
Melt temperature air shot	Thermal stability, overheating, nozzle/hot-tip mismatch	Must use safe purge procedure; not a cavity measurement
Microscopy / cross-section	Splay root cause, voids, inclusions, degradation particles	Often destructive; strong failure-analysis method
Polarized light	Residual stress in transparent amorphous polymers such as PC	Useful for clear parts; geometry dependent
X-ray / micro-CT	Internal voids, inclusions, short glass distribution, trapped air	Non-destructive but slower and higher cost
Ultrasonic / SAM / THz	Internal weld-line features, delamination, subsurface defects	Application-specific and needs expertise

Figure 9. Diagnostic methods. Use multiple evidence streams; visual inspection alone is insufficient for high-risk parts.

5.2 Visual inspection

- Use fixed lighting, fixed camera angle, fixed part orientation, and a severity scale. Uncontrolled lighting turns cosmetic inspection into opinion.
- Separate defects by location: near gate, flow-end, weld-line, thick section, runner/sprue, hot-tip residue, or random distribution.
- Drool and stringing should be photographed with a scale and noted by cycle state: after recovery, at mold open, during sprue pull, after carriage retract, or after robot takeout.
- Splay near the gate immediately after increasing decompression is a strong air-ingestion signal, but moisture must still be checked for hygroscopic materials.
- Black specks in hot runners require purge history, zone map, soak time, dead-spot inspection, and material residence review before blaming decompression alone.

5.3 Microscopy and destructive failure analysis

- Cross-section splay, bubbles, voids, cold slugs, gate freeze defects, and inclusions when visual inspection is ambiguous.
- Use microscopy to distinguish air bubbles from unmelted material, contamination, filler agglomerates, and degraded particles.
- For structural or medical components, visual cosmetic acceptance is not enough. Mechanical testing, MFR/MVR comparison, FTIR/DSC/GPC where available, and failure-analysis methods may be needed.

5.4 Mold-flow and barrel simulation

Mold-flow analysis is useful for predicting pressure demand, shear rate, flow-front behavior, air traps, weld lines, gate freeze, clamp force, and temperature distribution, but it does not automatically replace a shop-floor decompression DOE. Autodesk Moldflow material characterization uses a modified Cross-WLF viscosity model to characterize viscosity as a function of shear rate, temperature, and pressure [7]. Moldex3D documents suck-back behavior in barrel analysis and notes limitations, including that suck-back simulation is not supported in a model with hot runner attributes in the cited workflow [4].

Question	Simulation can help	Physical DOE still required
Will this gate create high shear or pressure loss?	Yes: shear/pressure prediction and gate comparison.	Yes: verify actual resin lot, machine response, and gate quality.
Will decompression eliminate drool?	Partially: barrel/nozzle pressure effect may be modeled in some tools.	Yes: drool/stringing depends on timing, nozzle contact, gate freeze, hot-runner state, and material behavior.
Will hot runner residence degrade material?	Simulation and hot-runner residence estimates can flag risk.	Yes: purge samples, MFR/MVR, color, specks, and mechanical properties must be checked.
Can viscosity curve replace decompression DOE?	No. In-mold rheology/viscosity curves measure flow resistance vs shear/fill speed.	Decompression DOE measures pressure relief and stability, not only viscosity.

5.5 Other methods that can match or exceed a basic decompression DOE

- Cavity pressure instrumented study: stronger than part-only evaluation because it shows actual in-cavity behavior, transfer repeatability, gate seal, and short-shot risk. Kistler notes that measuring pressure/temperature where molding occurs gives process transparency and supports shot-by-shot quality prediction [18].
- In-mold rheology / viscosity curve: useful when the suspected issue is flow resistance, material lot difference, gate shear, or temperature profile rather than decompression.
- Gate-freeze study: required when stringing or drool interacts with pack/hold and gate seal. Without gate-freeze data, decompression may be used to mask an overpacked or under-frozen process.
- Residence/degradation study: better than decompression DOE for black specks, odor, color shift, brittleness, MFR change, and material hang-up. Intertek describes capillary rheometry thermal stability testing where apparent melt viscosity is plotted versus residence time [6].
- Purge-and-hold challenge: intentionally hold material at temperature for a controlled time and compare purge samples, first shots, visual defects, and lab tests. This is critical for hot runners.
- Hot-runner balance audit: temperature actuals, valve pin timing, gate drops, pressure curves, fill balance, and cavity weights. This is often more relevant than screw pullback alone in hot runner molds.

6. Preventive Measures and Process Optimization

6.1 Material selection and preparation

- Choose a resin grade with a realistic flow window for the part wall thickness, flow length, gate size, and hot/cold runner architecture. Do not compensate for undersized gates by overheating material or overusing decompression.
- Use supplier-approved drying conditions and verify moisture where required. Covestro warns that moisture can create streaks/bubbles and hydrolysis-related molecular degradation that may not be visible externally [8].
- Control regrind percentage, fines, color concentrate carrier compatibility, and additive packages. Regrind often has extra thermal history and can shift viscosity or degradation sensitivity.

- For lubricants, flow promoters, and processing aids, validate downstream effects: plate-out, screw slip, hot-tip drool, vent contamination, part performance, regulatory compliance, and customer approval.
- For medical, pharmaceutical closure, food-contact, or automotive safety parts, material changes require formal validation and customer/regulatory review where applicable.

6.2 What the DOE tells you and how to optimize the process

DOE finding	Likely meaning	Actionable process response
Drool decreases as post-decompression increases, with no change in weight/cushion/splay.	Residual pressure was a real contributor.	Select the lowest effective setting; validate restart and production run duration.
Drool improves but splay appears.	Over-decompression or too-fast decompression may be pulling air. Moisture may also contribute.	Reduce amount or speed; verify dryer and nozzle seat; check front-zone/nozzle temperature.
No improvement in drool across decompression levels.	Problem may be nozzle temperature, hot-tip temperature, gate design, shutoff leak, back pressure, or material viscosity.	Stop increasing decompression. Audit temperatures, gate/tip, nozzle, and back pressure.
Part weight variation increases with decompression.	Shot volume recovery or check-ring behavior is destabilized.	Reduce decompression, inspect NRV/check ring, check screw recovery and cushion.
Black specks increase in hot runner at high decompression.	Possible oxygen/air ingestion, hot-tip dead spots, excessive residence, or degraded hang-up.	Reduce decompression; purge; inspect hot runner; review idle/startup temperature strategy.
Cold runner strings improve only when nozzle temperature is lowered.	Thermal/viscosity issue may dominate over pressure-relief issue.	Lock moderate decompression; optimize nozzle profile and sprue/gate freeze.
First-shot shortness after interruption worsens with pre-decompression.	Pre-decompression may reduce pressure readiness before injection.	Reduce or remove pre-decompression; validate startup and restart sequence.

6.3 Cold runner process optimization

- Start by stabilizing nozzle temperature and nozzle contact. A cold nozzle causes cold slugs; a hot nozzle can drool. Both can create stringing.
- Verify sprue bushing/nozzle radius and orifice compatibility. Mismatch can trap material, leak, or create drool unrelated to screw pullback.
- Run a gate-freeze study before final decompression. If hold pressure is active after gate freeze, it wastes pressure and can distort symptoms.
- Use decompression mainly to relieve nozzle/sprue pressure and prevent drool/stringing. Use cold slug wells and sprue/runner design to capture chilled material instead of relying on pullback alone.
- Observe sprue picker timing. If the picker pulls too early or too late, strings may appear even if the pressure-relief setting is correct.
- Avoid excessive carriage retract as a substitute for proper decompression. Carriage movement changes observation timing and may create nozzle-seat wear or material leakage issues.

6.4 Hot runner process optimization

- Treat hot-runner actual temperature, not just machine barrel temperature, as a DOE control variable. A one-zone drift can look like decompression failure.
- Document manifold, drop, and tip setpoints/actuals before every DOE run. Include soak time and whether the system was in production, idle, or startup mode.
- For thermal gates, evaluate gate freeze, drool, vestige, and tip temperature together. A hot tip may string; a cold tip may freeze off or cause short shots.
- For valve gates, lock valve sequence during the first DOE. If valve timing is suspected, study it separately or as a planned interaction factor.
- Be conservative with decompression. Hot runners are often more sensitive because the manifold/drops remain molten and can retain pressure or develop stagnant hot pockets.
- Confirm that decompression does not create color shift, black specks, or cavity imbalance after idle and restart. A steady-state clean sample is not enough for hot runner release.
- Use purge procedures that address hot tips and manifold dead spots. Black specks after long idle often require thermal management and purge changes, not only screw pullback changes.

6.5 Mold design improvements

Improvement	Cold runner value	Hot runner value
Optimize gate size/location	Reduces shear, improves fill balance, controls gate freeze and stringing.	Reduces tip shear, drool, gate blush, and cavity imbalance.
Uniform wall thickness	Reduces hesitation, flow lines, sink, differential pressure, and short-shot risk.	Same; also helps maintain consistent gate seal and thermal balance.
Cold slug wells	Captures chilled nozzle/sprue material before it reaches part.	Less applicable except hybrid systems; hot-tip cold slugs are handled by tip thermal design.
Vent design and maintenance	Prevents burns and false short-shot/splay conclusions.	Same; hot runner imbalance can move air-trap locations, so venting must match actual fill

		pattern.
Runner balance	Prevents cavity-to-cavity weight and fill variation in cold systems.	Manifold rheology/thermal balance must also be managed.
Nozzle/tip shutoff design	Can reduce drool but may create dead spots or maintenance needs.	Valve gates reduce visible drool but require pin timing, seal, and pressure control.

6.6 Machine calibration and maintenance

- Inspect screw, barrel, check ring, seat, nozzle tip, heater bands, thermocouples, and shutoff mechanisms on a preventive schedule. Do not use decompression to compensate indefinitely for worn hardware.
- Calibrate or verify screw position, injection pressure, back pressure, temperature zones, and recovery speed as part of process capability work.
- Trend recovery time, screw motor load, cushion, transfer pressure, and part weight. Decompression conclusions are stronger when machine signatures are stable.
- Use controlled purge and shutdown procedures. Material left hot in barrel, nozzle, hot tips, or manifold can invalidate the next DOE and create black-speck complaints.

6.7 In-process adjustments

During production, only make real-time adjustments inside the validated process window. The safe sequence is: verify material feed/drying and temperature actuals, confirm machine signatures, inspect visible drool/string timing, then adjust decompression in small increments. If a change fixes drool but creates weight drift, cushion drift, splay, specks, or short shots, back out the change and inspect root cause.

Adjustment	Use when	Do not use when
Increase post-decompression slightly	Drool/stringing occurs with otherwise stable fill, no splay, no shortness.	Splay/air defects already present or cushion unstable.
Reduce decompression amount	Splay, bubbles, short first shot, unstable cushion, or weight spread appears after pullback increase.	Clear residual pressure drool remains and no air/stability issue exists.
Slow decompression speed	Air draw suspected, especially near nozzle/hot tip.	Hydraulic/electric response is too slow to finish decompression before next event.
Adjust nozzle/front-zone temperature	Cold slug, drool, or stringing is temperature-driven.	Resin supplier max/min melt limits would be exceeded.
Adjust hot-tip/manifold temperature	Hot runner gate drool, freeze-off, or cavity imbalance is zone-specific.	No zone data exists; changing all zones blindly may create degradation.
Adjust back pressure	Melt density/mixing or drool pressure is linked to recovery.	Back pressure is being used to mask material, screw, or hot-runner faults.

6.8 Post-processing solutions

Post-processing should not be the primary remedy for decompression-related defects because drool, strings, cold slugs, splay, and degradation symptoms often indicate process or material risk. For cold runner systems, trimming, degating, tumble finishing, polishing, or painting can hide minor gate vestige or cosmetic residue but will not restore degraded molecular weight or fix voids. For hot runner systems, post-processing may remove vestige or surface blemishes, but black specks, color shift, and air-related defects usually require process correction, purge, or tooling work.

6.9 Redesign strategies

- Modify part geometry to remove abrupt flow restrictions, thick-to-thin transitions, and unnecessary flow splits that intensify shear and pressure loss.
- Move or resize gates to avoid long unsupported flow lengths, jetting, weld-line weakness, and hot-tip over-shear.
- Add cold slug wells in cold runner tools where chilled material enters the part.
- Redesign runner balance or hot-runner manifold layout if cavity-specific drool, stringing, or black specks persist despite stable processing.
- Collaborate with the mold maker when venting, gate land, hot-tip fit, valve-pin alignment, or nozzle/sprue geometry is outside process capability.

7. Industry Best Practices and Governance

- Use material supplier processing guides as hard constraints. Decompression, temperatures, drying, back pressure, and screw rpm must remain inside resin-safe windows unless the supplier approves otherwise.
- Use ASTM/ISO methods for lab work where material degradation is at issue. ASTM D3835 covers rheological properties at processing temperatures and shear rates and includes melt viscosity sensitivity/stability with respect to temperature and dwell time [5]. ASTM D1238 covers melt flow rate determination for thermoplastics and is often used for quality-control comparison of degraded vs undegraded resin [12].

- Use controlled cavity-pressure or machine-pressure data where risk justifies it. Pressure curves prevent "good-looking parts" from hiding unstable fill or transfer.
- Create separate setup sheets for cold runner and hot runner molds. Never copy decompression settings from a cold runner process into a hot runner without validation.
- For medical/pharmaceutical/automotive safety products, integrate decompression DOE into validation protocol, change control, and production part approval process where applicable.
- Train operators to report symptoms using timing and location, not vague labels. "String at cavity 7 thermal gate during mold open after 20 minutes idle" is actionable; "bad strings" is not.

7.1 Practical manufacturer-aligned recommendations

Source type	Relevant guidance	Shop-floor translation
Resin supplier processing guides	Eastman and Sumitomo examples caution that decompression should be minimal/small when needed because excessive suck-back can draw air into the nozzle and cause splay/blisters [2][3].	Find the minimum effective setting, not the maximum clean-looking setting.
Troubleshooting guides	BASF engineering thermoplastics guide links excessive melt temperature/residence/moisture with degradation and recommends reducing residence time in plasticizing unit and hot runner, drying material, and decompressing a few millimeters when material drips and cools before next shot [11].	Separate drool/cold slug correction from degradation correction; hot runner residence must be part of the review.
Simulation providers	Moldex3D supports suck-back behavior in barrel analysis under defined limitations and Autodesk Moldflow characterizes viscosity through shear/temperature/pressure models [4][7].	Use simulation to improve understanding, but prove final setting on press.
Sensor providers	Kistler emphasizes pressure/temperature measurement in the mold for transparency and shot-by-shot quality prediction [18].	Use sensors on high-cavity, medical, automotive, or high-scrap tools where part-only inspection is too slow or weak.

7.2 Peer-review release model

Because this document has not been independently peer-reviewed by named external experts, the best practice is to require local technical review before conversion into a controlled plant standard. At minimum, process engineering, tooling, quality, maintenance, and EHS should review the procedure. For critical parts, include the resin supplier, hot-runner supplier, machine supplier, or customer SQE as needed.

8. Emerging Technologies and Research

- Smart molding systems increasingly combine injection machine data, hot-runner controller data, cavity pressure, mold temperature, dryer data, and vision inspection. This supports faster root-cause isolation when drool, splay, or degradation signatures appear.
- In-mold sensors are mature enough for production monitoring. Cavity pressure and temperature sensors are especially useful for separating machine/nozzle behavior from actual cavity fill behavior.
- AI and machine-learning approaches are moving from parameter prediction toward in-mold-condition optimization. Recent research proposes using in-mold condition profiles and explainable AI to optimize injection molding quality, reducing dependence on subjective operator interpretation [22].
- Iterative learning control and digital twins are increasingly used to update process parameters cycle to cycle. For decompression DOE, the opportunity is not to let AI blindly change pullback, but to use data to detect when pressure relief, recovery, hot-runner balance, or moisture has drifted outside validated boundaries.
- Material innovations such as nano-additives, improved stabilizers, and self-healing polymers may widen processing windows in some applications, but they can also change viscosity, shear sensitivity, filler orientation, vent fouling, and hot-runner hang-up. Every material change must be validated.

9. Troubleshooting Guide and Safe Adjustment Path

The troubleshooting sequence below is designed to prevent the common mistake of chasing decompression before the process is stable. It is intentionally conservative. Safety, mold protection, and customer requirements override productivity pressure.

Step	Action	Pass/fail criteria	If fail
1	Confirm material identity, lot, drying, moisture, and feed consistency.	Material and drying match approved setup.	Correct material/dryer issue before DOE.
2	Stabilize barrel, nozzle, mold, and hot-runner actual temperatures.	Actuals are stable inside plant tolerance.	Fix heater, thermocouple, controller, water/oil circuit, or hot runner.

3	Verify baseline process: injection profile, transfer, hold, back pressure, screw rpm, cooling, cushion.	10-shot baseline is stable.	Do not run DOE; fix unstable baseline.
4	Inspect nozzle, sprue bushing, gate/tip, vents, check ring, and screw recovery.	No obvious mechanical defect.	Repair hardware or document known limitation.
5	Identify primary symptom timing and location.	Symptom is tied to cycle event and location.	Add photos/video with safe procedure.
6	Run decompression screening in small increments.	Drool/string improves without new defects.	If new air/splay/shortness occurs, reduce amount or speed.
7	For hot runner: validate cavity balance and restart.	All cavities pass after steady-state and restart.	Audit hot-runner zones, tip condition, valve timing, and purge.
8	Confirm by extended run.	Quality remains stable across production window.	Reopen root-cause investigation.
9	Lock and document.	Setup sheet, DOE data, photos, limits, and escalation rules complete.	Do not release as standard setup.

9.1 Symptom-based adjustment guide

Symptom	Likely decompression relation	Cold runner action	Hot runner action
Nozzle drool / ooze	Too little pressure relief, high nozzle temp, high back pressure, low viscosity.	Increase post-decompression in small steps; verify nozzle temperature and sprue bushing fit.	Check hot-tip/manifold pressure and temperature; increase cautiously if no air/splay.
Sprue/gate stringing	Too little pressure relief, gate not frozen, nozzle too hot, takeout timing.	Confirm gate freeze and sprue puller timing; adjust post-decompression and nozzle temp.	Check thermal gate temp, valve timing, gate land, and cavity-specific imbalance.
Splay after decompression change	Too much or too-fast pullback, air ingestion, moisture.	Reduce amount/speed; check dryer and nozzle seat.	Reduce amount/speed; check tip seals, manifold leaks, and moisture.
Short first shot	Too much pre/post decompression, low pressure recovery, tip freeze, NRV issue.	Reduce decompression; verify cushion and check ring.	Reduce decompression; check hot-tip temperature and valve timing.
Black specks / odor	Not a normal decompression target; may be degradation, oxygen, dead spots, hot runner residence.	Purge barrel/nozzle; reduce melt residence/temp; inspect screw/check ring.	Purge manifold/tips; reduce idle temperatures; inspect dead spots and zone drift.
Unstable cushion / weight	Over-decompression, check-ring leakage, recovery instability.	Reduce decompression; run NRV check.	Same plus check cavity balance and manifold pressure recovery.
Cold slug near gate	Nozzle too cold, drool cools then injects, excessive decompression creating void/cooling at nozzle.	Optimize nozzle temp, cold slug well, sprue design; avoid over-pull.	Check hot-tip freeze-off and drop temperature; do not over-pull.

9.2 Common pitfalls

- Changing decompression, nozzle temperature, back pressure, screw rpm, hot-runner zones, and cooling at the same time.
- Calling the best drool setting “optimum” without checking splay, first-shot fill, part weight, cushion, and restart behavior.
- Running a hot runner DOE without recording actual tip/manifold temperatures and soak time.
- Using a cold runner setup sheet for a hot runner mold or vice versa.
- Ignoring material dryness and calling moisture splay a decompression issue.
- Ignoring worn check rings and calling cushion drift a decompression issue.
- Using excessive decompression to hide a bad nozzle seat, damaged hot tip, poor gate design, blocked vent, or unstable back pressure.
- Failing to purge degraded material before DOE, especially after idle, color change, or hot-runner temperature excursion.
- Declaring a result from one or two shots. Use replicate shots and confirmation runs.
- Using decompression changes outside validated bounds on regulated or customer-controlled processes without change approval.

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Appendix A. Shop-Floor Data Sheet

Field	Record
Date / shift / technician	
Machine ID / screw diameter / barrel capacity	
Mold ID / cavitation / cold or hot runner / gate type	
Resin grade / lot / color / regrind %	
Dryer temperature / dew point / moisture reading	
Barrel temps actual / nozzle temp actual	
Mold temp actual / water or oil circuit notes	
Hot runner controller zones setpoint and actual	
Baseline decompression values and timing	
DOE factor levels	
Shot count per condition	
Drool length / string length method	
Part weights by cavity	
Cushion readings	
Visual rating method and lighting	

Splay / blisters / black specks / odor notes	
Pressure curve source and file name	
Selected setting and rationale	
Restart validation results	
Approvals	

Appendix B. Peer-Review and Release Checklist

Reviewer role	Required review items	Approval criteria
Process engineering	DOE design, factor ranges, controls, data analysis, final window.	All process conclusions are supported by measured data.
Tooling engineering	Gate, runner, hot runner, venting, nozzle/sprue compatibility, wear.	Tool condition does not invalidate the decompression setting.
Quality engineering	Measurement method, sample size, acceptance criteria, traceability, customer requirements.	Data package supports release and audit needs.
Maintenance	Screw, check ring, barrel, nozzle, heater bands, hot-runner hardware, shutoffs.	Machine/tool hardware is capable and documented.
Material/resin supplier	Drying, melt temperature, residence, shear, additives, regrind, purge.	Process remains inside resin-safe processing window.
EHS / safety	Purge, hot material, fume, lockout/tagout, observation method, robot/sprue picker safety.	Procedure can be executed safely by trained personnel.
Production leadership	Standard work, training, setup sheet control, escalation rules.	Operators and technicians can repeat the process.

End of manual - use with plant-specific safety procedures, resin supplier guidance, customer requirements, and validated process control documentation.